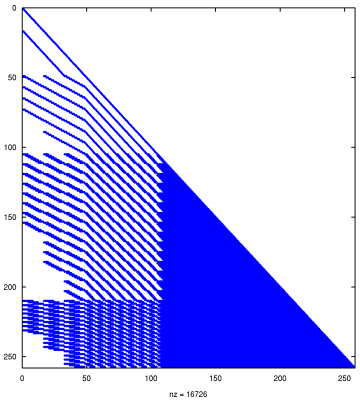
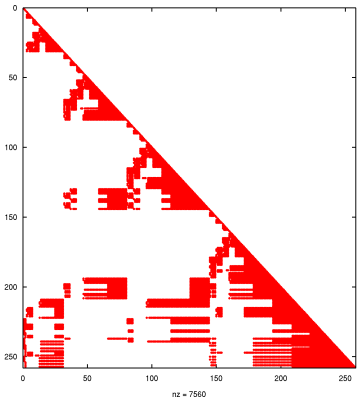
MA385 Part 3: Linear Algebra 1

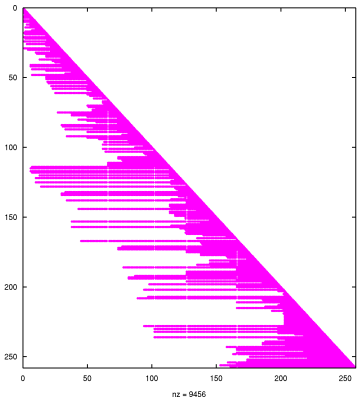
3.2: Gaussian Elimination

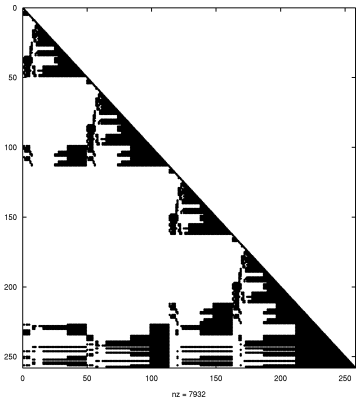
Dr Niall Madden

02 November 2026 (W9.1)









1. Outline of Section 3.2

1. Gaussian Elimination
2. Row operations are matrix multiplication
3. Triangular Matrices
   - Lower Triangular
   - Upper Triangular
   - Properties (I)
   - Matrix partitioning
   - Properties (II)
4. Exercises

For more, see Section 2.2 of text:
https://ebookcentral.proquest.com/lib/nuig/reader.action?docID=221072&ppg=51&c=UERG



Carl Freidrich Gauß, Germany, 1777-1855. Although he produced many very important original ideas, Gaussian Elimination wasn't one of them. The Chinese knew this algorithm about 2000 years ago. His actual contributions included major discoveries in the areas of number theory, geometry, numerical analysis and astronomy.

2. Gaussian Elimination

Gaussian Elimination is an **exact** method for solving linear systems. It is "exact" in two senses:

- We preform an exact fixed number of operations and we arrive at the solution. That different from our **iterative methods** for nonlinear equations.
- At the end, we arrive at the exact/true at the end of the process (ignoring round-off error). This is in contrast to **approximate** methods studied earlier in the module.

Note: There are approximate methods for solving linear systems, but they are not part of this module, unfortunately.

Example 3.2.1

Consider the problem:

$$\begin{pmatrix} -1 & 3 & -1 \\ 3 & 1 & -2 \\ 2 & -2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -9 \end{pmatrix}$$

We can perform a sequence of elementary row operations to yield the system:

$r_2 \leftarrow r_2 + 3r_1$

$$\begin{pmatrix} -1 & 3 & -1 \\ 0 & 10 & -5 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ -5 \end{pmatrix}.$$

$-3(-1, 3, -1) + (3, 1, -2) = (0, 10, -5)$

2. Gaussian Elimination

The first steps in detail:

$$A = \begin{pmatrix} -1 & 3 & -1 \\ 3 & 1 & 2 \\ 2 & -2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ -9 \end{pmatrix}$$

$$r_2 + 3r_1 \Rightarrow \begin{pmatrix} -1 & 3 & -1 \\ 0 & 10 & -1 \\ 2 & -2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ -9 \end{pmatrix}$$

This is equivalent to

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 3 & -1 \\ 3 & 1 & 2 \\ 2 & -2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \\ -9 \end{pmatrix}$$

3. Row operations are matrix multiplication

Gaussian Elimination: perform elementary row operations such as

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

being replaced by

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} + \mu_{21} a_{11} & a_{22} + \mu_{21} a_{12} & a_{23} + \mu_{21} a_{13} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} =$$

$$A + \mu_{21} \begin{pmatrix} 0 & 0 & 0 \\ a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \end{pmatrix}$$

where $\mu_{21} = -a_{21}/a_{11}$, so that $a_{21} + \mu_{21} a_{11} = 0$.

3. Row operations are matrix multiplication

Note that

$$\begin{pmatrix} 0 & 0 & 0 \\ a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

so we can write the row operation as $\left(I + \mu_{21} E^{(21)}\right) A$, where $E^{(pq)}$ is the matrix of all zeros, except for $e_{pq} = 1$.
In general each of the row operations in Gaussian Elimination can be written as

$$\left(I + \mu_{pq} E^{(pq)}\right) A \quad \text{where } 1 \leq q < p \leq n, \tag{1}$$

and $\left(I + \mu_{pq} E^{(pq)}\right)$ is an example of a **Unit Lower Triangular Matrix**.

3. Row operations are matrix multiplication

We can conclude that each step of the process will involve multiplying A by a unit lower triangular matrix, resulting in an upper triangular matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \times & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \times & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ \times & 1 & 0 \\ \times & \times & 1 \end{pmatrix} A \sim \begin{pmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \end{pmatrix}$$

Definition 3.2.1 (Lower Triangular)

$L \in \mathbb{R}^{n\times n}$ is a *Lower Triangular (LT) Matrix* if the only non-zero entries are on or below the main diagonal, i.e., if $l_{ij} = 0$ for $1 \le i < j \le n$.
It is a *unit Lower Triangular matrix* if, in addition, $l_{ii} = 1$.

Examples: $L_2 = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$ is LT. It is Unit LT if $a = d = 1$.

$L_3 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$ is LT (but not unit LT)

The Identity matrix is unit LT. $\to p > q$
The zero matrix, & $E^{(pq)}$ are LT.

Definition 3.2.2 (Upper Triangular)

$U \in \mathbb{R}^{n\times n}$ is an *Upper Triangular (UT) matrix* if $u_{ij} = 0$ for $1 \le j < i \le n$. It is a *Unit Upper Triangular Matrix* if $u_{ii} = 1$.

Examples:

$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ is UT. It is Unit LT if $a = d = 1$

$$U = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$ is UT

$u_{21} = u_{31} = u_{32} = u_{41}, \ldots, u_{43} = 0$

Triangular matrices have many important properties. A very important one is: **the determinant of a triangular matrix is the product of the diagonal entries**.

We'll leave the proof of this as an exercise (see Exer 3.2.2).

However, it tells us two important facts:

- A triangular matrix has an inverse if, and only if, none of the diagonal entries are zero.
- A unit triangular matrix has a determinant of 1.

Eg $L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}$ $\det(L) = l_{11} l_{22} l_{33}$

There are other important properties of triangular matrices, but first we need the idea of **matrix partitioning**.

Definition 3.2.3 (Submatrix)

X is a *submatrix* of A if it can be obtained by deleting some rows and columns of A.

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ -1 & -2 & -2 & 5 \\ 0 & 1 & 5 & 9 \\ -3 & -1 & 0 & 2 \end{pmatrix}$$

Submatrices include (1), $\begin{pmatrix} 1 & 2 \\ -1 & -2 \end{pmatrix}$, $\begin{pmatrix} -2 & -2 & 5 \\ 1 & 5 & 9 \\ -1 & 0 & 2 \end{pmatrix}$

$\begin{pmatrix} 2 & 4 \\ -2 & 5 \\ 1 & 9 \end{pmatrix}$ But not $\begin{pmatrix} 1 & 2 \\ -2 & -2 \\ 5 & 9 \end{pmatrix}$

Definition 3.2.4 (Leading Principal Submatrix)

The **Leading Principal Submatrix** of order k of $A \in \mathbb{R}^{n\times n}$ is $A^{(k)} \in \mathbb{R}^{k\times k}$ obtained by deleting all but the first k rows and columns of A. (Simply put, it's the $k \times k$ matrix in the top left-hand corner of A).

Example:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$A^{(1)} = (1)$$

$$A^{(2)} = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$$

$$A^{(3)} = A$$

Matrix partitioning

To **partition a matrix** means to divide it into contiguous blocks that are submatrices.

Example:

$$A = \left(\begin{array}{cc|c} a & b & c \\ d & e & f \\ \hline g & h & i \end{array}\right)$$ is partitioned

as $A^{(2)} = \begin{pmatrix} a & b \\ d & e \end{pmatrix}$ $B = \begin{pmatrix} c \\ f \end{pmatrix}$

$C = \begin{pmatrix} g & h \end{pmatrix}$ $D = \begin{pmatrix} i \end{pmatrix}$

$$A = \left(\begin{array}{c|c} A^{(2)} & B \\ \hline C & D \end{array}\right)$$

Finished here Mon 2/11.

Next suppose that if A and V are matrices of the same size, and each are partitioned

$$A = \left(\begin{array}{c|c} B & C \\ \hline D & E \end{array}\right), \qquad \text{and} \qquad V = \left(\begin{array}{c|c} W & X \\ \hline Y & Z \end{array}\right),$$

where B is the same size as W, C is the same size as X, etc. Then

$$AV = \left(\begin{array}{c|c} BW + CY & BX + CZ \\ \hline DW + EY & DX + EZ \end{array}\right).$$

Theorem 3.2.1 (Properties of Lower Triangular Matrices)

For any integer $n \geq 2$:

(i) If L_1 and L_2 are $n \times n$ *Lower Triangular* (LT) Matrices that so too is their product $L_1 L_2$.

(ii) If L_1 and L_2 are $n \times n$ *Unit Lower Triangular* matrices, then so too is their product $L_1 L_2$.

(iii) L_1 is nonsingular if and only if all the $l_{ii} \neq 0$. In particular all Unit LT matrices are nonsingular.

(iv) The inverse of a LT matrix is an LT matrix. The inverse of a unit LT matrix is a unit LT matrix.

We restate Part (iv) as follows:

If L^{-1} exists then it is LT

Suppose that $L \in \mathbb{R}^{n\times n}$ is a lower triangular matrix with $n \geq 2$, and that there is a matrix $L^{-1} \in \mathbb{R}^{n\times n}$ such that $L^{-1}L = I_n$. Then L^{-1} is also a lower triangular matrix.

Proof The proof is by induction. First, let's suppose that $n = 2$....

Write $L = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$. since $\det(L) = ad$ we know $ad \neq 0$.

Then $L^{-1} = \frac{1}{ad}\begin{pmatrix} d & 0 \\ -c & a \end{pmatrix}$. This is a lower triangular matrix, as required.

Next, let's suppose that L is an $n \times n$ LT matrix, and that the result holds for any $(n-1) \times (n-1)$ LT matrix. Since L is triangular, and is invertable, and all its diagonal entries are non-zero, since $\det(L) = \prod_{j=1}^{n} l_{jj} \neq 0$.

This means $L^{(n-1)}$, its leading principle submatrix of order $n-1$ also has an inverse (why?).

See board!

Now partition L by its last row and column: $L = \left(\begin{array}{c|c} L^{(n-1)} & \mathbf{0} \\ \hline \mathbf{c}^T & d \end{array}\right)$,

where here $\mathbf{0}$ is the $(n-1)$ vector of zeros, $\mathbf{c}^T$ is some $(n-1)$ row vector, and d is a scalar.

We know L^{-1} exists and we partition it as $L^{-1} = \left(\begin{array}{c|c} W & \mathbf{x} \\ \hline \mathbf{y}^t & z \end{array}\right)$,

where here $\mathbf{x}$ and $\mathbf{y}$ are $(n-1)$ vectors, and z is some scalar.
Let I_k denote the $k \times k$ identity matrix. Then, $I_n = LL^{-1}$ gives

$$\left(\begin{array}{c|c} I_{n-1} & \mathbf{0} \\ \hline \mathbf{0}^T & 1 \end{array}\right) = \left(\begin{array}{c|c} L^{(n-1)} & \vec{\mathbf{0}} \\ \hline \mathbf{c}^T & d \end{array}\right) \left(\begin{array}{c|c} W & \mathbf{x} \\ \hline \mathbf{y}^T & z \end{array}\right) = \ldots$$

$$= \left(\begin{array}{c|c} L^{(n-1)}W + \vec{0}\mathbf{y}^T & L^{(n-1)}\mathbf{x} + \vec{0}y \\ \hline \mathbf{c}^T W + d\mathbf{y}^T & \mathbf{c}^T\mathbf{x} + dz \end{array}\right)$$

To show that L^{-1} is LT, we need

(1) W to be LT

(2) $\vec{x} = \vec{0}$

For (1), we have $L^{(n-1)} W = I_{n-1}$

So $W = \left(L^{(n-1)}\right)^{-1}$. Thus W is UT by our inductive hypothesis.

For (2), we have $L^{(n-1)} \vec{x} + \vec{0} y = \vec{0}$

So $\vec{x} = \left(L^{(n-1)}\right)^{-1} \vec{0} = \vec{0}$,

as required.

Theorem 3.2.2 (Properties of Upper Triangular Matrices)

Statements that are analogous to those concerning the properties of lower triangular matrices hold for upper triangular and unit lower triangular matrices. (For proof, see the exercises at the end of this section).

5. Exercises

Exercise 3.2.1

Every step of Gaussian Elimination can be thought of as a left multiplication by a unit lower triangular matrix. That is, we obtain an upper triangular matrix U by multiplying A by k unit lower triangular matrices: $L_k L_{k-1} L_{k-2} \dots L_2 L_1 A = U$, where each $L_i = I + \mu_{pq} E^{(pq)}$, and $E^{(pq)}$ is the matrix whose only non-zero entry is $e_{pq} = 1$. Give an expression for k in terms of n.

Exercise 3.2.2

Let $L = (l_{ij})$ be a lower triangular $n \times n$ matrix. Show that $\det(L) = \prod_{j=1}^{n} l_{jj}$.

Hence or otherwise, also show that if $U = (u_{ij})$ is an upper triangular matrix, then $\det(U) = \prod_{j=1}^{n} u_{jj}$.